

Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained in Python and HTML/CSS files.
2	Proper File Structure	Files and folders are logically organized	Yes	Organized into templates, static, model, and dataset folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as temperature, humidity, prediction, model are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are clearly named based on their purpose
5	Code Comments	Inline and block comments explain logic	Partial	Comments are added in important sections only.
6	Modular Design	Code is split into reusable functions/m	Yes	Data preprocessing, prediction, and UI modules are separated.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated code has been minimized.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation and exception handling are implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Model Training & Prediction	High
2	Machine Learning Model	Scikit-learn	Crop Recommendation	High
3	Prediction Function	Python	Flask Web Application	High
4	User Interface	HTML, CSS	Web Application	High
5	Flask Backend	Flask, Python	Request Processing & Prediction	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules.
Readability	5	Code is easy to understand with meaningful names and formatting
Reusability	5	Core modules can be reused in future agricultural AI projects.
Documentation / Comments	4	Important sections are documented; additional comments can improve maintainability.
Overall Score	4.8/5	The project follows good coding standards, modular design, and is easy to maintain and extend.